34721 Linear Control Design 1 Exam F25

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:

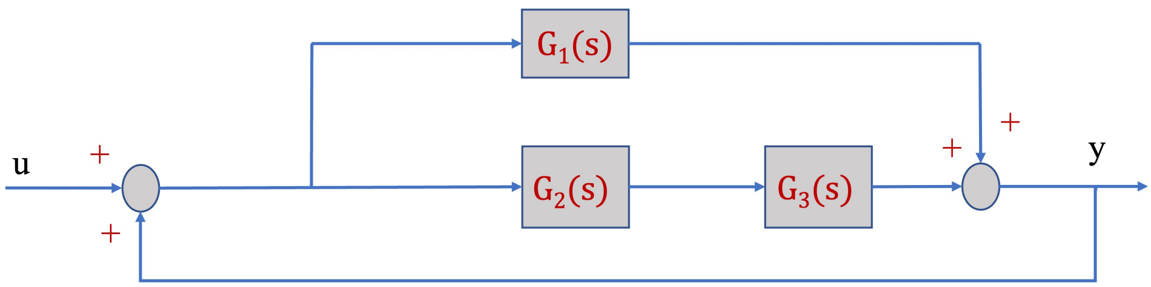
- Der er altid netop ét svar som er mere rigtigt end de andre
- Studerende kan kun vælge ét svar per spørgsmål
- Hvert rigtigt svar giver 1 point
- Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"

- There is always only one correct answer – a response that is more correct than the rest
- Students are only able to select one answer per question
- Every correct answer corresponds to 1 point
- Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

Page 1

Question 1 Consider the system with input $u(t)$ and output $y(t)$ in the diagram below where the first-order linear systems have the following transfer functions $G_1(s) = \frac{4}{s+10}, G_2(s) = \frac{1}{s+2}, G_3(s) = \frac{s+2}{s+10}$

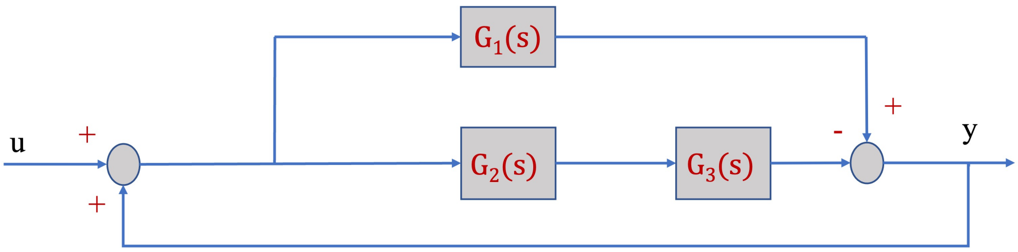


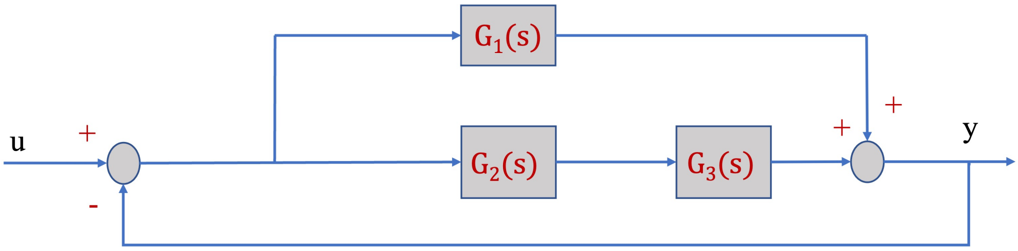
Calculate the $y(0)$ and the $\lim_{t\to\infty} y(t)$ when $u(t) = e^{-2t} + \theta(t)$, where $\theta(t)$ is the unit step function and $t >= 0$.

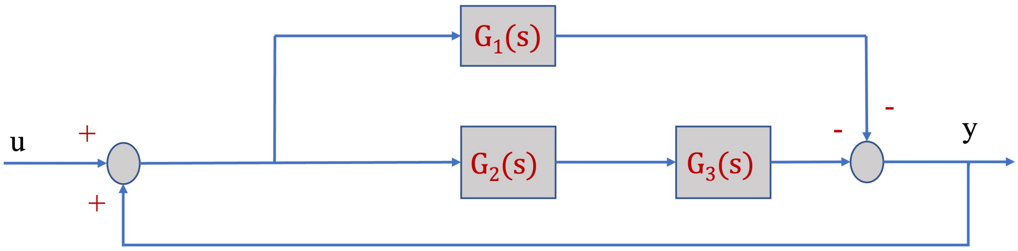
- [x] $y(0) = 0, \lim_{t\to\infty} y(t) = 1$
- [ ] $y(0) = 0, \lim_{t\to\infty} y(t) = 0.33$
- [ ] $y(0) = 1, \lim_{t\to\infty} y(t) = 1$
- [ ] $y(0) = 1, \lim_{t\to\infty} y(t) = \infty$

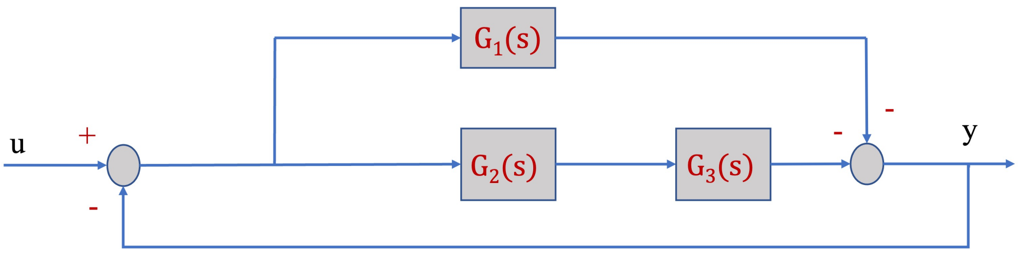
Question 2 Select the correct block diagram that corresponds to the I/O Transfer Function $H(s) = \frac{Y(s)}{U(s)} = \frac{3}{s+7}$.

$G_1(s) = \frac{4}{s+10}, G_2(s) = \frac{1}{s+2}, G_3(s) = \frac{s+2}{s+10}$

- [x] 

- [ ] 

- [ ] 

- [ ] 

Question 3 Consider the transfer function

$$F(s) = \frac{Y(s)}{U(s)} = \frac{3(s+2)(s+4)}{(s+3)(s^2+6s+6k+8)}$$

For which range of parameter k the system with input $u(t)$ and output $y(t)$ is asymptotically stable?

- ○ $k \in \mathbb{R}$
- ✔ $k > -\frac{4}{3}$
- ○ $k \leq -1$
- ○ $k < 0$
- ○ $k < -\frac{2}{3}$

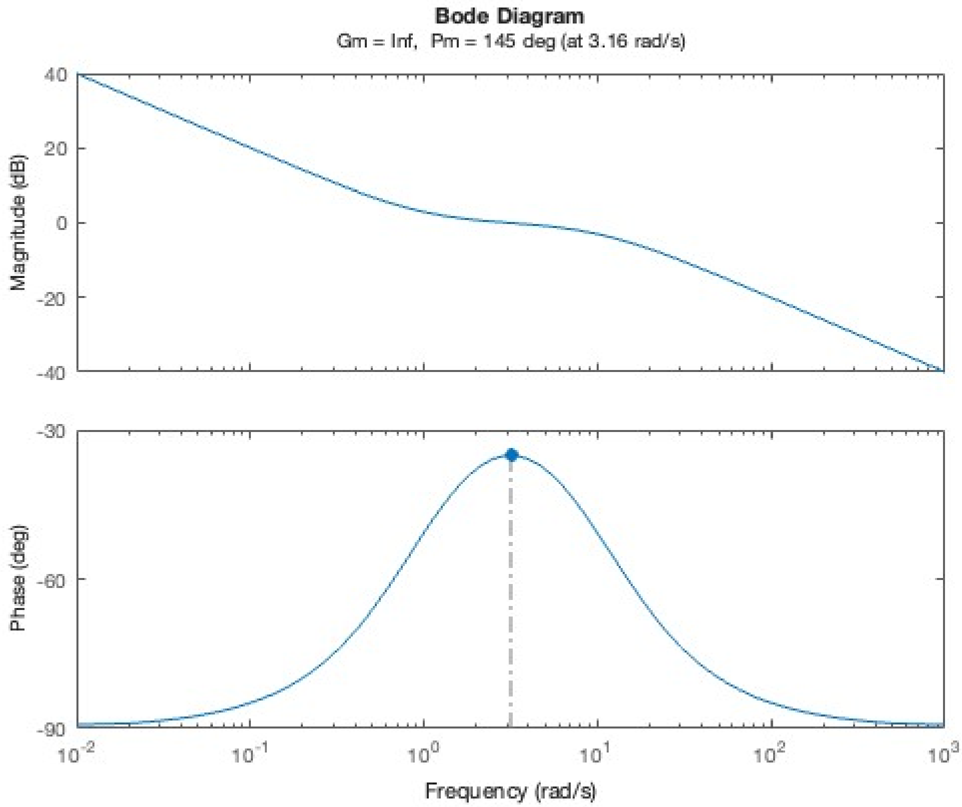
Question 4 A unity-feedback closed-loop control system has open-loop transfer function

$$G(s) = \frac{K}{s(s+3)}$$

Which one of the following statements is correct?

- ○ The system has finite steady-state error for a step input and infinite error for ramp and parabolic inputs.
- ○ The system has infinite steady-state error for all standard inputs.
- ✔ The system has zero steady-state error for a step input.
- ○ The system has zero steady-state error for step and ramp inputs and finite error for a parabolic input.

Question 5 The Bode diagram shown corresponds to a transfer function:



Which of the following statements about the poles and zeros of the system is most consistent with the diagram?

- [x] The system has a pole at the origin, a zero at $\omega = 1$, and a pole at $\omega = 10$
- [ ] The system has a zero at the origin, a pole at $\omega = 1$, and another pole at $\omega = 10$
- [ ] The system has a single pole at $\omega = 10$, and no zeros
- [ ] The system has two poles at the origin and a zero at $\omega = 1$

Question 6 The system has the following transfer function:

$G(s) = 8\frac{(s+2)}{(s+0.2)(s^2+10s+25)}$

For a unit step input, approximately how long does it take for the output to settle within 1% of its final value?

- 0.2
- 5
- ✔ 25
- 12.5
- 2.5

Question 7 Linearize the given nonlinear rotational mass-spring-damper system described by the following differential equation around the working point ($\theta_0 = 2\mathrm{rad}$, $\dot{\theta}_0 = 4\mathrm{rad/s}$):

$$J\ddot{\theta}(t) + b\theta(t)^3 + d\sqrt{\dot{\theta}(t)} = T(t)$$

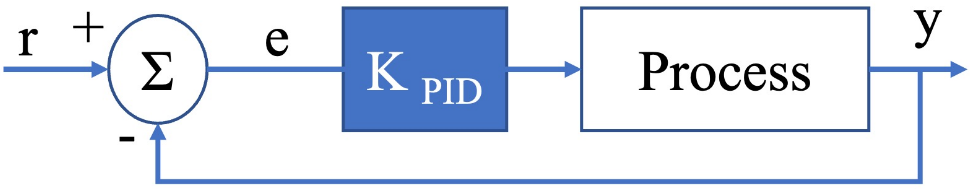
where the input is torque $T(t)$ and the output is angular velocity $\dot{\theta}(t)$. The constants are: $J = 5\mathrm{kg{\cdot}m^2}$, $b = 3\mathrm{N{\cdot}m/rad^3}$, $d = 2\mathrm{N{\cdot}m{\cdot}s^{1/2}/rad^{1/2}}$, $\theta_0 = 2\mathrm{rad}$, and $\dot{\theta}_0 = 4\mathrm{rad/s}$.

- ✔ $\Delta T = 5\Delta\ddot{\theta} + 36\Delta\theta + 0.5\Delta\dot{\theta}$
- $\Delta T = 5\Delta\ddot{\theta} + 24\Delta\theta + \Delta\dot{\theta}$
- $\Delta T = 5\Delta\ddot{\theta} + 36\Delta\theta + \Delta\dot{\theta}$
- $\Delta T = 5\Delta\ddot{\theta} + 24\Delta\theta + 0.5\Delta\dot{\theta}$

Question 8 Consider the PID-controlled system in the figure. Apply Ziegler-Nichols tuning rule for the determination of the values of parameters K_p, T_i and T_d. The closed-loop transfer function is $\frac{Y(s)}{R(s)} = \frac{K_p}{s(s+1)(s+5)+K_p}$

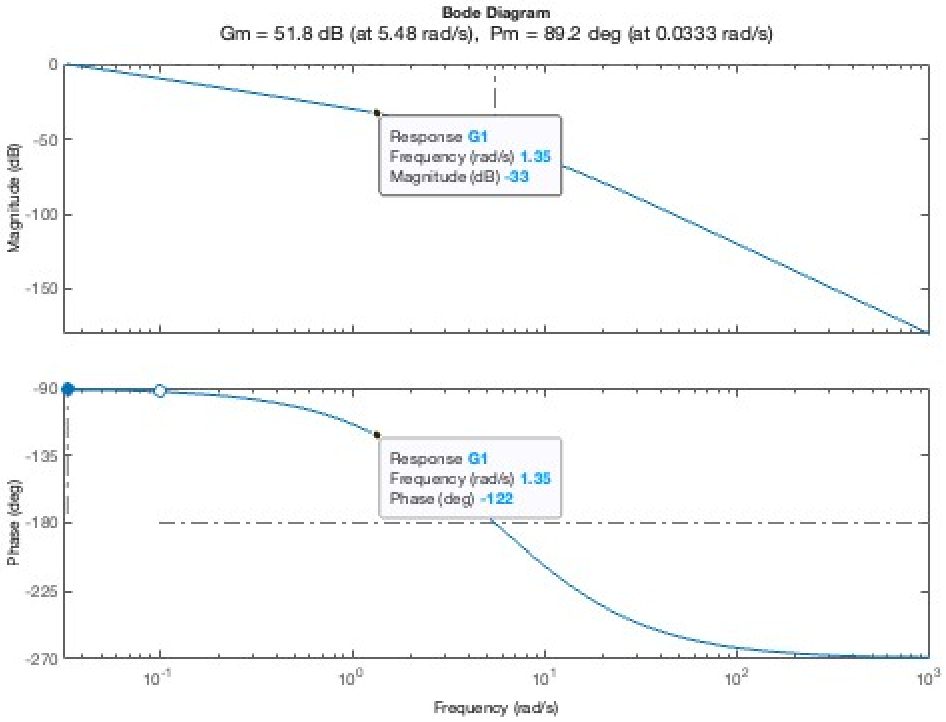
The period of the sustained oscillation is $P_u = \frac{2\pi}{\omega}$ where ω is the frequency of the sustained oscillation when the critical gain $K_u = 30$.

Select the correct transfer function of the PID controller.



- [x] $K_{PID}(s) = 18(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 15(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 30(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 18(1 + \frac{1}{2.809s} + 0.35124s)$

Question 9 The Bode diagram of the open-loop transfer function $G_1(s)$ shows the following stability margin values:
Gain margin: $G_m = 51.8\,\mathrm{dB}$
Phase margin: $P_m = 89.2^\circ$



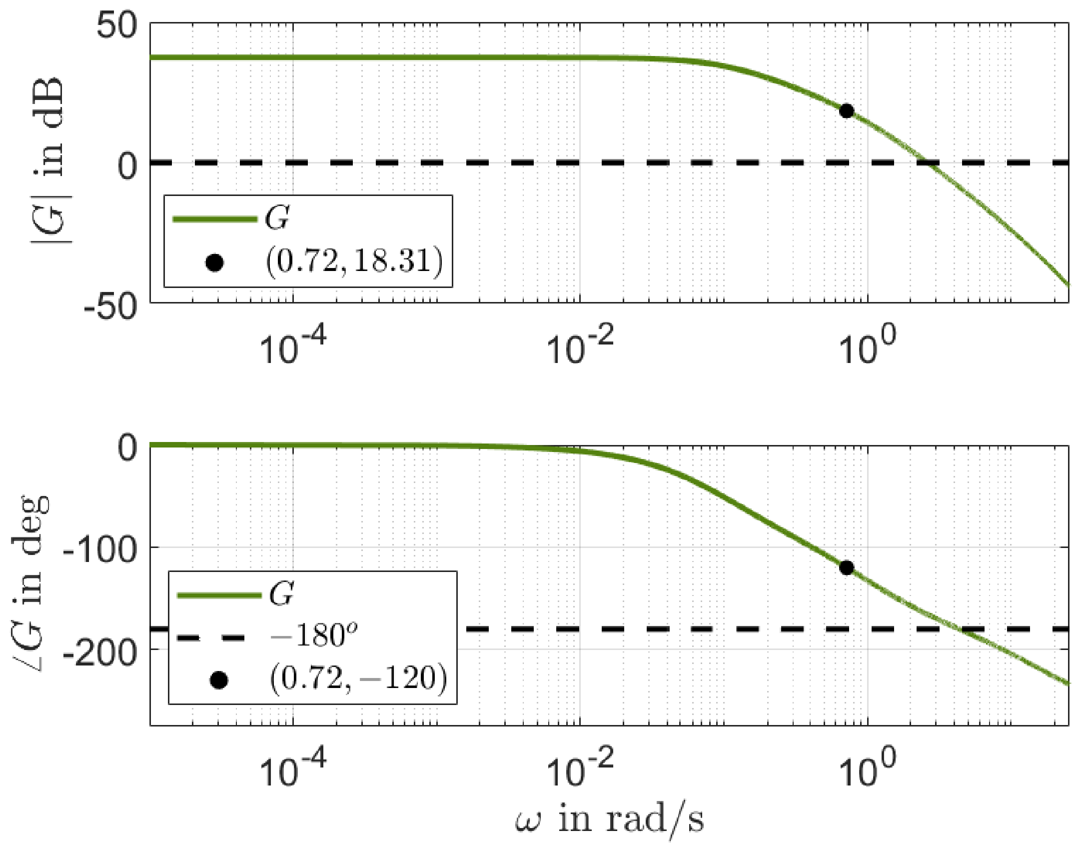
You wish to adjust the proportional gain K_p so that the phase margin is approximately 60°. What will be the new gain margin after this adjustment?

- ○ The new gain margin is approximately 33 dB
- ○ The gain margin remains unchanged if the phase at -180° does not move
- ○ The new gain margin is greater than 51.8 dB
- ○ The new gain margin is equal to 51.8 dB
- ✔ The new gain margin is less than 51.8 dB

Question 10 Which of the following statements about Bode diagrams is true?

- ○ A pole in the transfer function introduces a phase lead of $+90^{\circ}$
- ✔ The magnitude plot of a Bode diagram for a first-order system with a pole at $\omega = a$ will change slope from 0 to $-20dB/decade$ at $\omega = a$
- ○ The phase shift caused by a pole or zero begins exactly at the break frequency
- ○ A zero in the transfer function contributes a decrease of 20 dB/decade to the magnitude plot starting at its break frequency.

Question 11. Consider a system described by a transfer function $G(s)$ with open-loop Bode plot as shown in the following figure.



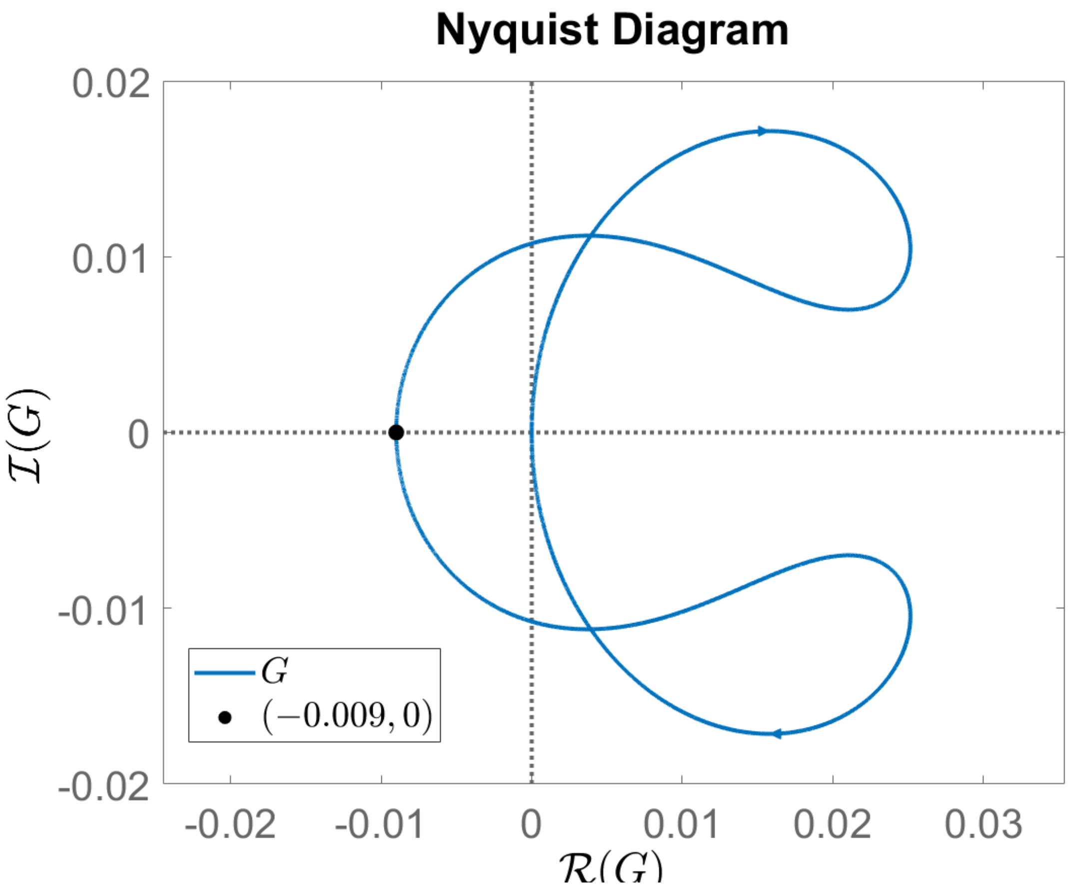
A P-controller needs to be designed so that a phase margin $\gamma_M = 60^o$ is achieved. The proportional gain K_P is approximately:

- ○ $K_P = 18.31$
- ○ $K_P = 0.9$
- ○ $K_P = -18.31$
- ○ $K_P = 2.7$
- ✔ $K_P = 0.12$

Question 12. Which of the following statements is correct?

- [x] When identifying the transfer function of an input rate/amplitude-limited system it is better to use small step inputs to produce input/output data.
- [ ] An input amplitude-limited (saturated) system is always linear.
- [ ] Increasing the value of K_a in a feedback-based anti-windup strategy for an input amplitude-limited system decreases the undershoot in the closed-loop step response.
- [ ] Increasing the proportional gain K_P in an input amplitude-limited (saturated) system which is controlled by a PI-Lead controller alleviates (reduces) the effect of integrator wind-up.
- [ ] When designing controllers for an input rate-limited system, the crossover frequency should be chosen as far as possible from the dominant pole of the system.

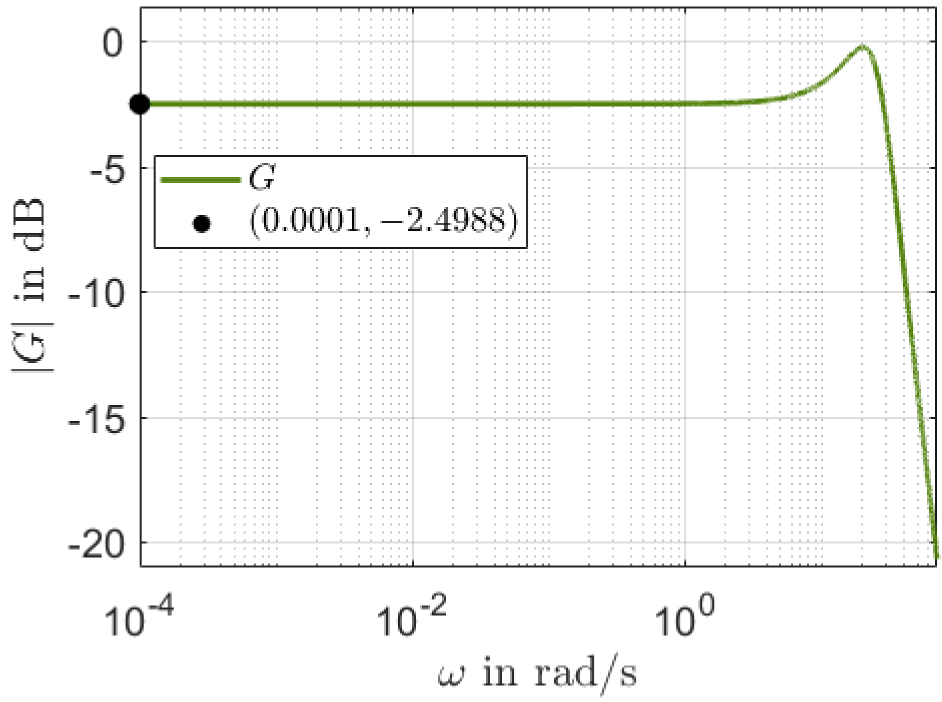
Question 13. Consider a system described by a transfer function $G(s)$ with one pole on the right half complex plane. The associated open-loop Nyquist plot is shown in the following figure.



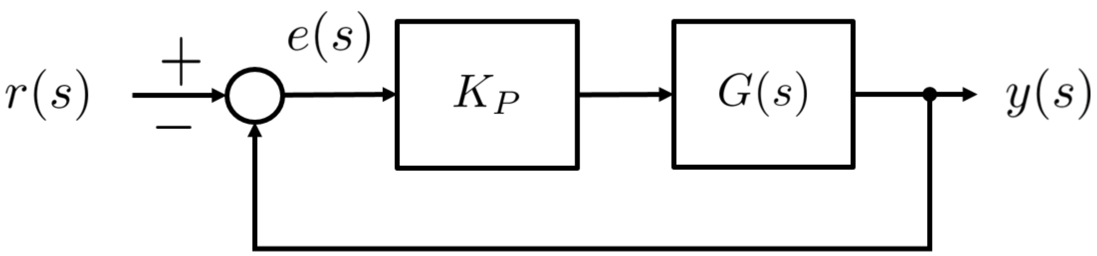
The Nyquist plot intersects the real axis along the counter-clockwise direction at the point $(-0.009, 0)$. A P-controller with $K_P = 115$ is applied to the system and the loop is closed with unit feedback. Which of the following statements is the correct one?

- [x] The closed-loop system is stable
- [ ] The closed-loop system is unstable.
- [ ] The closed-loop system is marginally stable.
- [ ] The system retains the same stability properties in closed-loop (as in open-loop).
- [ ] The open-loop system is stable.

Question 14. Consider a system described by a transfer function $G(s)$ with a magnitude Bode plot as shown in the figure below:



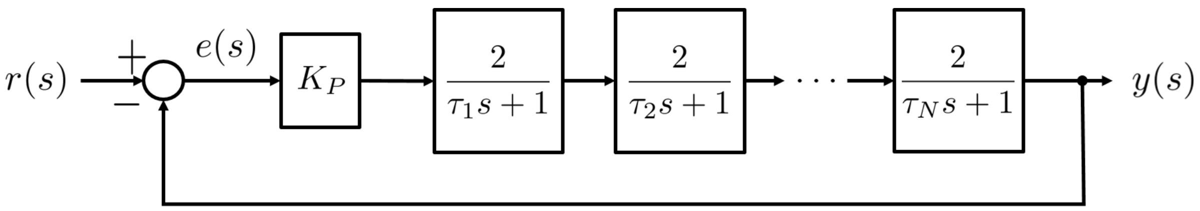
A P-controller with gain K_P is applied to the system and the loop is closed with unit feedback as in the following figure.



If the steady state error $e_{SS} = \lim_{s \to 0} e(s)$ due to a unit step change in the reference signal is $e_{SS} = 0.1$, the proportional gain K_P is approximately:

- ○ $K_P = 0.4$
- ○ $K_P = 1$
- ○ $K_P = 2.5$
- ✔ $K_P = 22.5$
- ○ $K_P = 7.95$

Question 15. Consider a system described by a transfer function $G(s)$ that consists of $N = 9$ low-pass filters in series as shown in the block diagram below:



If the steady state unit step response of the closed-loop system is $y_{SS} = 0.999$, then K_P is approximately:

- [x] $K_P = 1.95$
- [ ] $K_P = 1$
- [ ] $K_P = 18$
- [ ] $K_P = 25$
- [ ] $K_P = 5.8$

Question 16. A system is described by the following transfer function

$$G(s) = \frac{14}{s^3 + 30s^2 + 257s + 2}$$

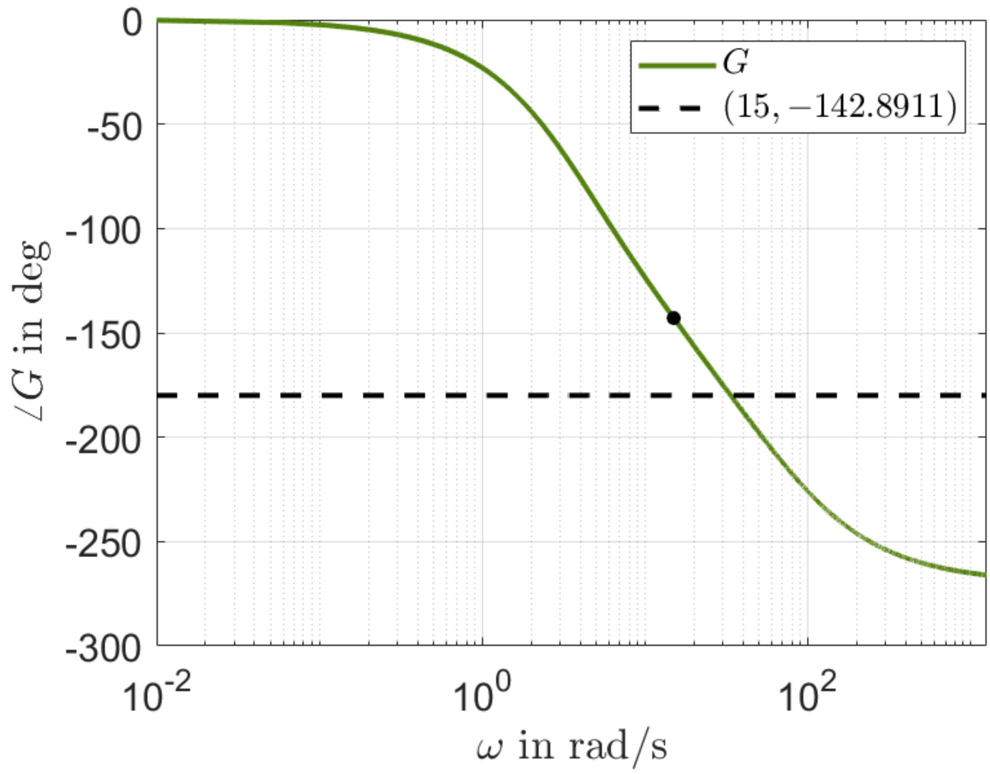
A PI-Lead controller

$$C(s) = K_P \underbrace{\frac{\tau_i s + 1}{\tau_i s}}_{C_{PI}(s)} \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{C_D(s)}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 75^o$. If ω_c is the open-loop crossover frequency, $N_i = \omega_c \tau_i = 3$ and $\alpha = 0.05$, the controller proportional gain K_P is approximately:

- [ ] $K_P = 34$
- [ ] $K_P = 235.9$
- [ ] $K_P = -34$
- [x] $K_P = 50$
- [ ] $K_P = 0.02$

Question 17. Consider the system described by a transfer function with the following phase Bode plot:



A P-Lead-**Lag** controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$

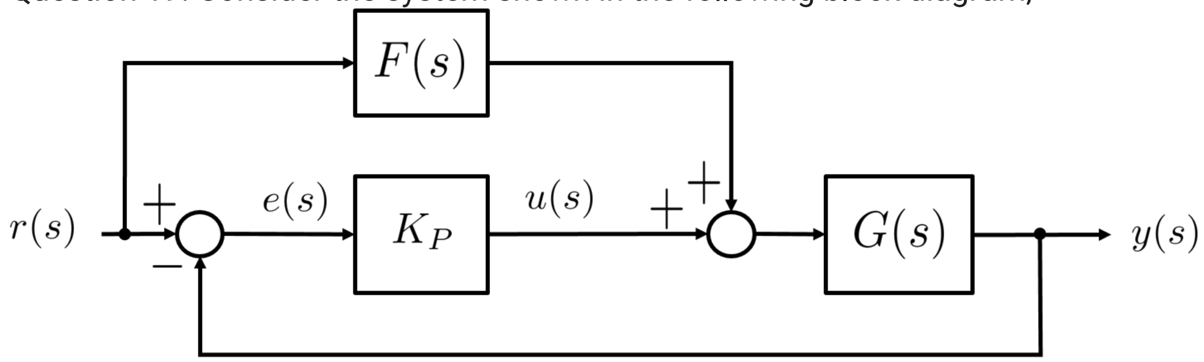
with $\alpha = 0.2$ and $N_i = 5$ needs to be designed for the system. The new crossover frequency (after applying the controller) is selected as $\omega_c = 15$ rad/s. If a phase margin $\gamma_M = 75^o$ is desired for the open-loop system $C(s)G(s)$, what is the (approximate) value of the parameter β of the Lag-part?

- [x] $\beta = 1.54$
- [ ] $\beta = -2.5$
- [ ] $\beta = 0.17$
- [ ] $\beta = 0.58$
- [ ] $\beta = 0$

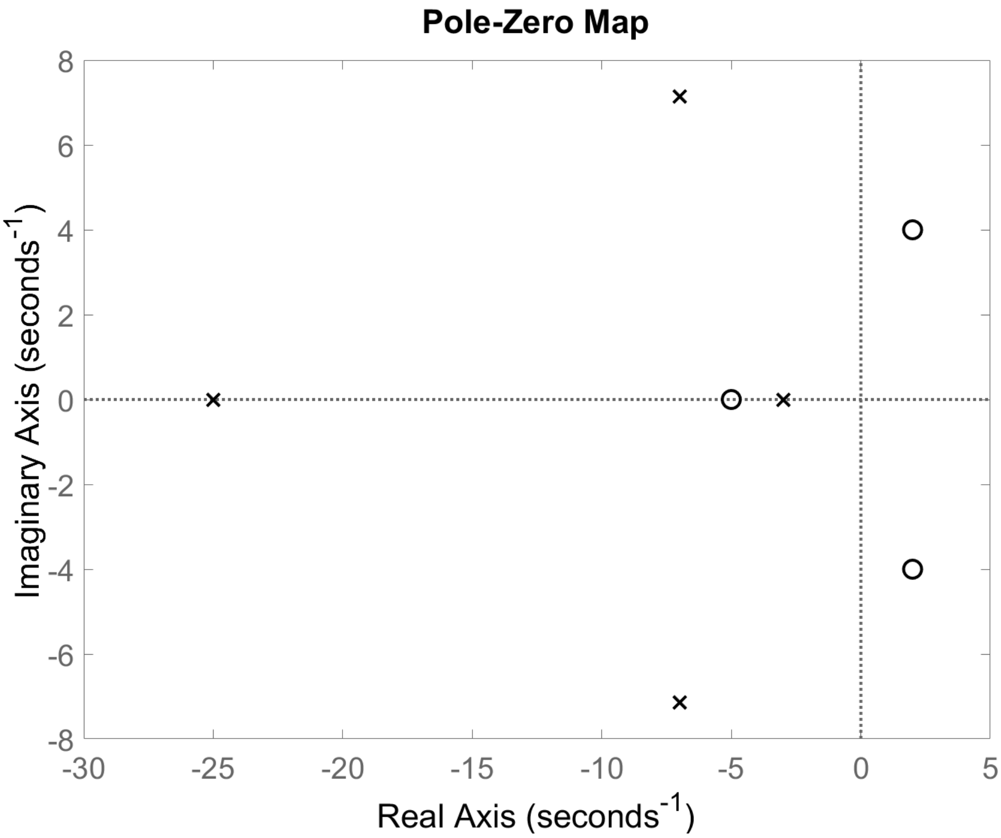
Question 18. Which of the following statements is correct?

- [x] A closed-loop system with 0 dB gain at very low (theoretically 0) frequencies implies zero steady-state error.
- [ ] Pre-filters (reference filters) are used to stabilise open-loop unstable systems.
- [ ] Increasing the proportional gain K_P of a P-controller will shift the phase curve in the open-loop Bode plot upwards.
- [ ] Encirclement of the $(-1, 0)$ point on a Nyquist diagram always implies instability of the unit-feedback closed-loop system.
- [ ] In a PI-controller design, there can be a value N_i for which the I-part gives positive phase contribution at the new crossover frequency.

Question 19. Consider the system shown in the following block diagram,



where $K_P > 0$ and the transfer function $G(s)$ has the following zeros-poles diagram:



A feed-forward controller $F(s)$ needs to be designed such that the steady-state error is zero. Which of the following is an appropriate design for $F(s)$?

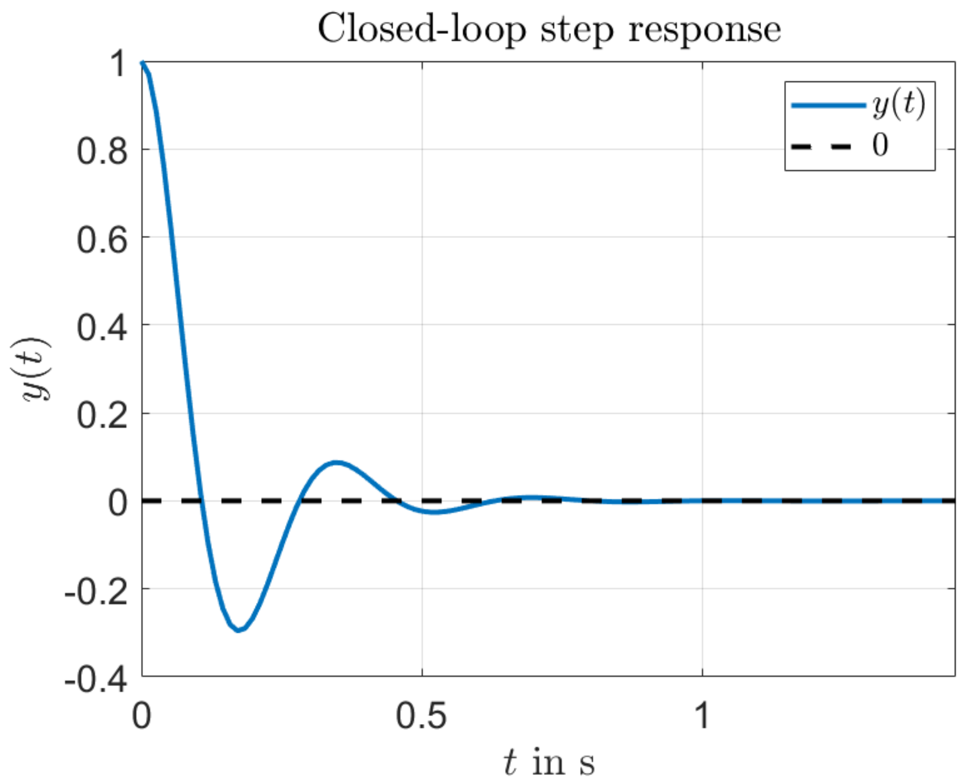
- [ ] $F(s) = G(0)$
- [ ] $F(s) = \dfrac{1}{G(s)}$
- [x] $F(s) = \dfrac{1}{G(s)(\tau_f s + 1)}, \ \tau_f = 0.01$
- [ ] $F(s) = \dfrac{K_P G(s)}{\tau_f s + 1}, \ \tau_f = 0.01$

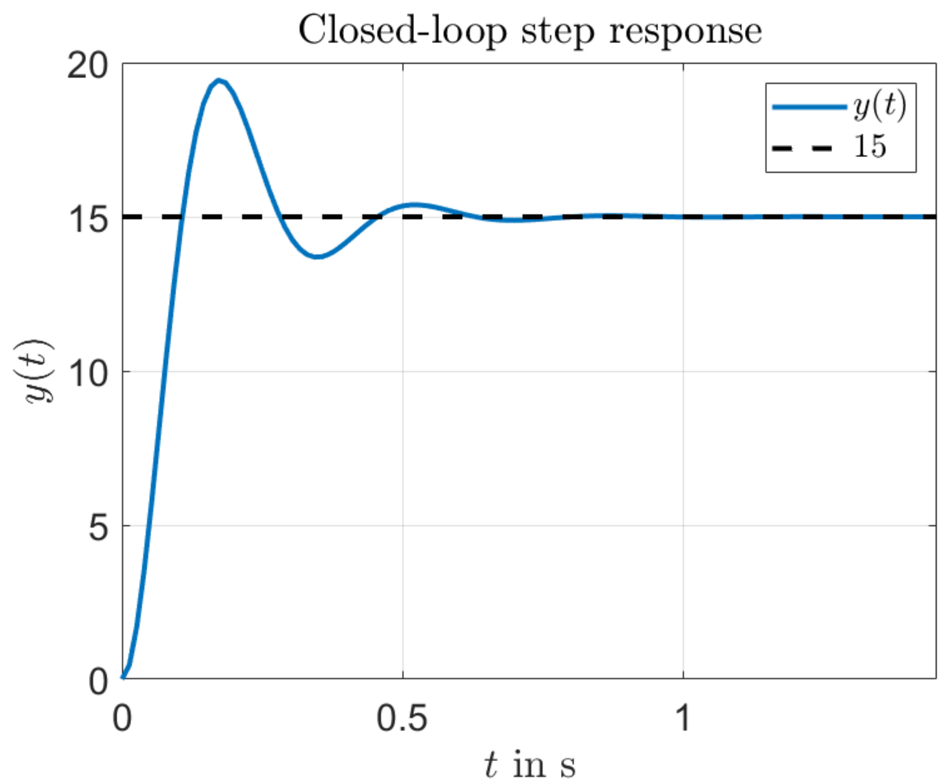
- [ ] $F(s) = \dfrac{1}{G(0)}$

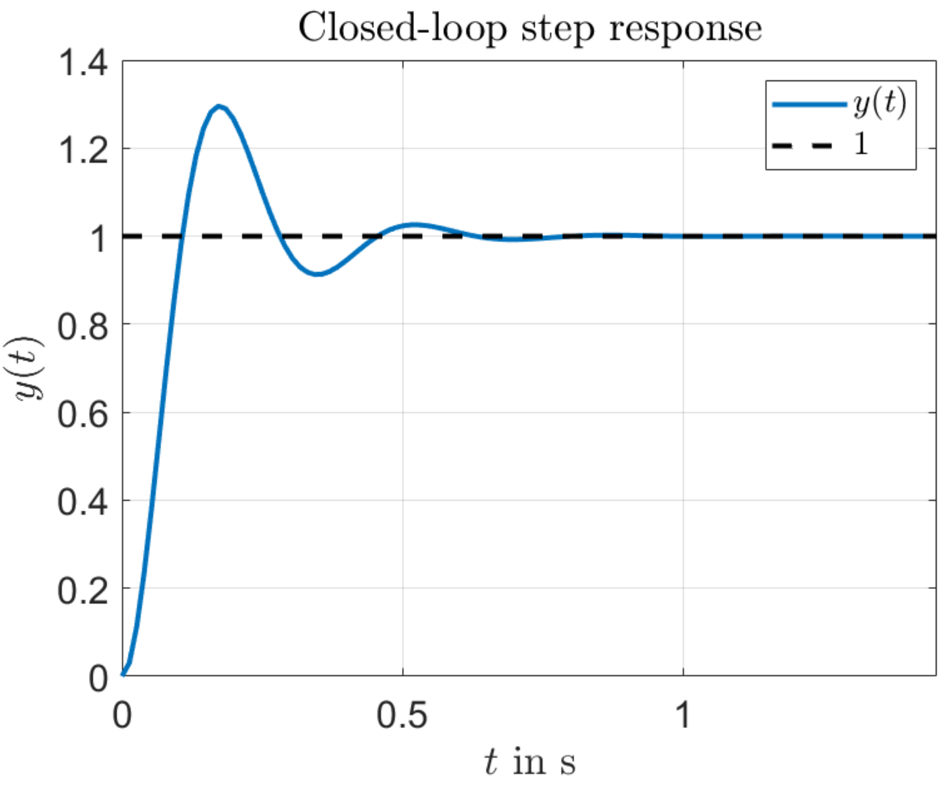
Question 20. Consider the system described by the following differential equation

$$\ddot{y}(t) + 14\dot{y}(t) = 25u(t)$$
$$u(t) = K_P(r(t) - y(t))$$
$$r(t) = \begin{cases} 1 \text{ if } t > 0 \\ 0 \text{ elsewhere} \end{cases}$$

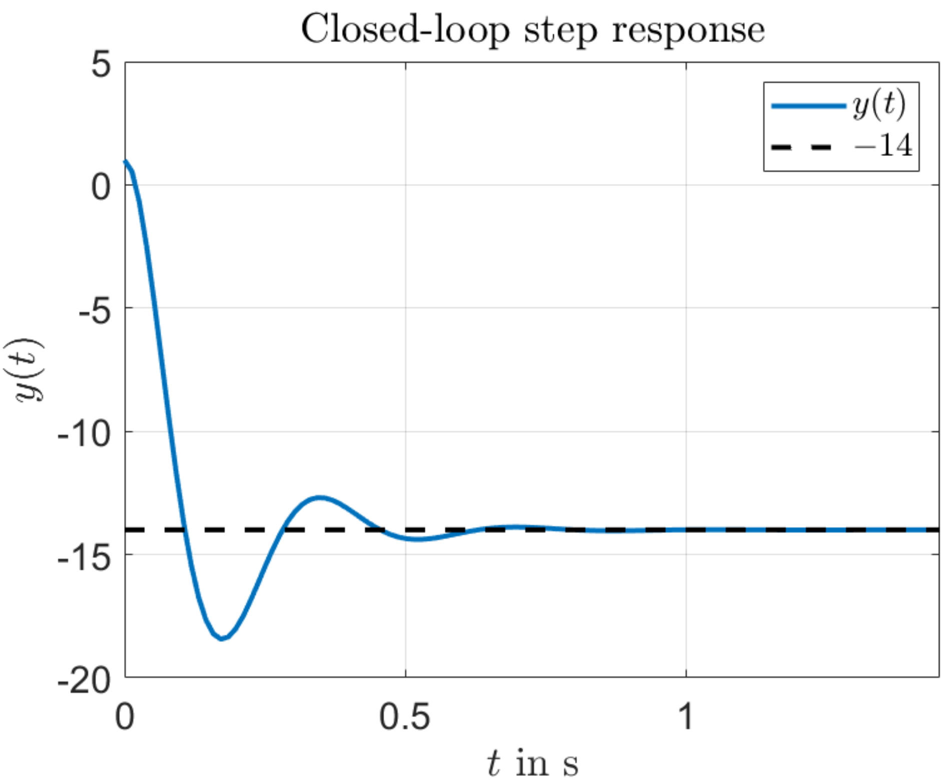
where $y(t)$ is the output of the system, $u(t)$ is the control input and $r(t)$ is a unit step reference signal, respectively. With $K_P = 15$ and assuming zero initial conditions, i.e. $y(0) = \dot{y}(0) = \ddot{y}(0) = u(0) = r(0) = 0$, which of the following plots represents the step response of the closed-loop system?





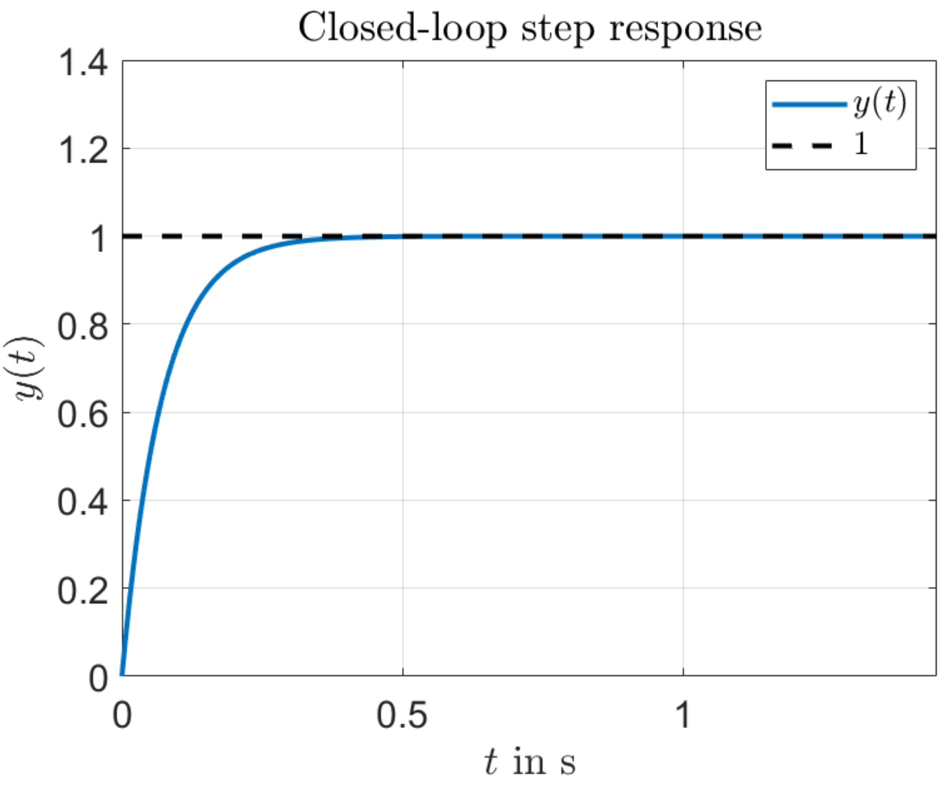


Closed-loop step response
$y(t)$
1
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
t in s



Closed-loop step response
$y(t)$
-14
5
0
-5
-10
-15
-20
0
0.5
1
t in s

○



Closed-loop step response
y(t)
1
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
y(t)
t in s